## EE3621: Digital Signal Processing | III B.Tech EEE

### Faculty Reference Document — Textbook Replacement

## PREFACE FOR FACULTY

Direct form realizations of high-order IIR filters suffer from catastrophic **coefficient quantization sensitivity**: A tiny change in high-order polynomial coefficients causes large displacements of roots, often shifting poles outside the unit circle and causing instability. **Cascade realization** factors the transfer function into a product of Second-Order Sections (SOS / biquads), isolating pole pairs and guaranteeing stability.

**Pedagogical Strategy:**

1. Demonstrate polynomial root sensitivity: Why 16-bit direct form filters fail for orders $N \geq 4$.
2. Formulate the Cascade structure: $H(z) = g\prod_{k=1}^{K} H_k(z)$.
3. Master **Optimal Pole-Zero Pairing Rules**: Pair complex conjugate poles with the closest transmission zeros in the $z$-plane to minimize section peak-to-average gain.
4. Master **Optimal Section Ordering Rules**: Order sections in sequence of increasing $Q$-factor (increasing pole radius $r \to 1$) to maximize dynamic range and minimize register overflow.
5. Formulate $L_2$ and $L_\infty$ scaling between sections.

## 1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Factor** high-order IIR transfer functions into second-order biquad sections (SOS).
2. **Apply** pole-zero pairing rules to minimize frequency response peaking.
3. **Order** biquad sections to optimize signal-to-noise ratio (SNR) and prevent overflow.
4. **Draw** complete Cascade realization signal flow graphs.

## 2. MATHEMATICAL FOUNDATIONS

### 2.1 The Cascade Realization Formulation

$$H(z) = g\prod_{k=1}^{K} H_k(z) = g\prod_{k=1}^{K} \frac{b_{k0} + b_{k1}z^{-1} + b_{k2}z^{-2}}{1 + a_{k1}z^{-1} + a_{k2}z^{-2}}, \quad K = \lceil N/2 \rceil$$

Each biquad $H_k(z)$ realizes one real pole pair and one real zero pair:

$$1 + a_{k1}z^{-1} + a_{k2}z^{-2} = (1 - p_k z^{-1})(1 - p_k^* z^{-1}) = 1 - 2r_k\cos(\theta_k)z^{-1} + r_k^2 z^{-2}$$

### 2.2 Pairing and Ordering Rules

1. **Pole-Zero Pairing:**
   - Pair the complex pole pair $p_k$ having highest $Q$ (closest to unit circle) with the zero pair $z_k$ closest to it in angle $\theta$.
   - This cancels large resonance peaks, keeping the section gain $|H_k(e^{j\omega})|$ relatively flat.
2. **Section Ordering:**
   - Arrange sections in order of **increasing pole radius** $r_k$ (increasing $Q$ / peaking).
   - Low-$Q$ sections first $\rightarrow$ High-$Q$ sections last.
   - Prevents premature signal saturation in early stages.

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## 3. WORKED NUMERICAL EXAMPLES

### Example 18.1: 4th-Order IIR Cascade Realization

**Problem:** Realize $H(z) = \frac{(1+z^{-1})(1-\sqrt{2}z^{-1}+z^{-2})}{(1-0.9e^{j\pi/4}z^{-1})(1-0.9e^{-j\pi/4}z^{-1})(1-0.6z^{-1})(1+0.5z^{-1})}$ in Cascade Form with optimal pairing.

**Solution:**

1. **Poles:**
   - Complex pair: $p_{1,2} = 0.9e^{\pm j\pi/4} \implies 1 - 2(0.9)\cos(\pi/4)z^{-1} + 0.81z^{-2} = 1 - 1.2728z^{-1} + 0.81z^{-2}$. (Radius $r = 0.9$, high $Q$).
   - Real poles: $p_3 = 0.6, ; p_4 = -0.5 \implies (1 - 0.6z^{-1})(1 + 0.5z^{-1}) = 1 - 0.1z^{-1} - 0.3z^{-2}$. (Low $Q$).
2. **Zeros:**
   - Complex pair: $z_{1,2} = 1 \cdot e^{\pm j\pi/4} \implies 1 - \sqrt{2}z^{-1} + z^{-2} = 1 - 1.4142z^{-1} + z^{-2}$ (at angle $\pi/4$).
   - Real zeros: $z_3 = -1 \implies (1 + z^{-1})$.
3. **Optimal Pairing:**
   - Pair poles at angle $\pi/4$ ($p_{1,2}$) with zeros at angle $\pi/4$ ($z_{1,2}$).
   - Section 1 (Low $Q$): $H_1(z) = \frac{1+z^{-1}}{1-0.1z^{-1}-0.3z^{-2}}$.
   - Section 2 (High $Q$): $H_2(z) = \frac{1-1.4142z^{-1}+z^{-2}}{1-1.2728z^{-1}+0.81z^{-2}}$.
4. **Ordering:** Section 1 followed by Section 2.

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## 4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

### Question 1 (15 Marks)

**(a)** Why are direct form structures not suitable for implementing high-order IIR filters? Explain coefficient quantization sensitivity. *(6 Marks)* **(b)** Realize the transfer function in Cascade Form:

$$H(z) = \frac{1 + \frac{1}{3}z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{4}z^{-1} + \frac{1}{2}z^{-2}\right)}$$

Draw the complete signal flow graph using Direct Form II biquads. *(9 Marks)*

**Model Answer & Step-by-Step Marking Rubric:**

- **Part (a):**
  - Root sensitivity explanation: Roots of $A(z)$ depend non-linearly on polynomial coefficients. In high-order polynomials, $1$ coefficient perturbation can displace poles by $50$ or move them outside $|z| = 1$ *(6 Marks)*
- **Part (b):**
  - Split into 1st-order and 2nd-order sections: $H_1(z) = \frac{1+\frac{1}{3}z^{-1}}{1-\frac{1}{2}z^{-1}}, \quad H_2(z) = \frac{1}{1-\frac{1}{4}z^{-1}+\frac{1}{2}z^{-2}}$ *(4 Marks)*
  - Neatly drawn Cascade SFG of $H_1(z)$ cascaded into $H_2(z)$ using canonical DF-II sections *(5 Marks)*

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## 5. PYTHON VERIFICATION SCRIPT

```
import scipy.signal as signal

b = [1, 1/3]
a = np.convolve([1, -0.5], [1, -0.25, 0.5])
sos = signal.tf2sos(b, a)
print("Computed SOS:")
print(sos)
```